Chapter B5: The human body – staying alive

Overview

From previous study, learners should appreciate that cells work together in multi-cellular organisms – in a hierarchy of cells, tissues, organs and systems – to support the functioning of each cell and of the organism as a whole. This chapter develops understanding of how cells and systems work together to support life in the human body.

In Topic B5.1, learners consider how the substances essential for chemical reactions are transported into, out of and around the human body, and why exchange surfaces are necessary. In Topics B5.2 and B5.3 they explore how the nervous and endocrine systems help the body to detect and respond to external and internal changes. Topic B5.4 illustrates the importance of maintaining a constant internal environment.

The essential role of hormones in human reproduction is explored in Topic B5.5, followed in Topic B5.6 by consideration of what can happen when certain structures and systems – including the regulation of blood sugar, structures in the eye and neurons in the nervous system – go wrong.

Learning about the human body before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- appreciate the hierarchical organisation of multicellular organisms: from cells to tissues to organs to systems to organisms
- be able to identify, name, draw and label the basic parts of the human body
- have a basic understanding of the function of muscles
- be familiar with the tissues and organs of the human digestive system, including adaptations to function
- understand the basic structures and functions of the gas exchange system in humans, including adaptations to function
- understand the mechanism of breathing to move air in and out of the lungs, and be able to use a pressure model to explain the movement of gases
- understand, in outline, how nutrients and water are transported within animals, including humans
- be able to identify and name the main parts of the human circulatory system
- be familiar with the functions of the heart, blood vessels and blood
- know which part of the body is associated with each sense.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

B5.1 How do substances get into, out of and around our bodies?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
Oxygen, water and molecules from food are essential for chemical reactions in cells in the human body, including cellular respiration and synthesis of biomass. Carbon dioxide and urea are waste products that need to be removed from cells before they reach toxic levels. Moving these substances into, around and out of the body depends upon interactions between the circulatory, gaseous exchange, digestive and excretory systems. Oxygen and carbon dioxide diffuse between blood in capillaries and air in alveoli. Water and dissolved food molecules are absorbed from the digestive system into blood in capillaries. Waste products including carbon dioxide and urea diffuse out of cells into the blood. Urea is filtered out of the blood by the kidneys into urine. Partially-permeable cell membranes regulate the movement of these substances; gases move across the membranes by diffusion, water by osmosis and other substances by active transport. The heart, blood vessels, red blood cells and plasma are adapted to transport substances around the body.	1. describe some of the substances transported into and out of the human body in terms of the requirements of cells, including oxygen, carbon dioxide, water, dissolved food molecules and urea 2. explain how the partially-permeable cell membranes of animal cells are related to diffusion, osmosis and active transport 3. describe the human circulatory system, including its relationships with the gaseous exchange system, the digestive system and the excretory system 4. explain how the structure of the heart is adapted to its function, including cardiac muscle, chambers and valves 5. explain how the structures of arteries, veins and capillaries are adapted to their functions, including differences in the vessel walls and the presence of valves 6. explain how red blood cells and plasma are adapted to their functions in the blood	**Practical work:** • dissect lamb's heart to observe atria, ventricles and valves • investigate valves in an arm vein (tourniquet around bicep; when veins become prominent, gently try to push blood in each direction)
To sustain all the living cells inside humans and other multi-cellular organisms, exchange surfaces increase the surface area:volume ratio, and the circulatory system moves substances around the body to decrease the distance they have to diffuse to and from cells.	7. explain the need for exchange surfaces and a transport system in multicellular organisms in terms of surface area:volume ratio 8. calculate surface area:volume ratios M1c, M5c	**Practical work:** • investigate the effect of surface area:volume ratio on diffusion of dye into agar cubes

B5.2 How does the nervous system help us respond to changes?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
In order to survive, organisms need to detect and respond to changes in their external and internal environments. The highly adapted structures of the nervous system facilitate fast, short-lasting responses to stimuli. In a stimulated neuron, an electrical impulse passes along the axon. Most axons have a fatty sheath to increase impulse transmission speed. An impulse is transmitted from one neuron to another across a synapse by the release of transmitter substances, which diffuse across the gap and bind to receptors on the next neuron, stimulating it.	1. explain how the components of the nervous system work together to enable it to function, including sensory receptors, sensory neurons, the CNS, motor neurons and effectors 2. explain how the structures of nerve cells and synapses relate to their functions ⓘ Learners are not expected to explain nerve impulse transmission in terms of membrane potentials	
Reflexes provide rapid, involuntary responses without involving a processing centre, and are essential to the survival of many organisms. In some circumstances the brain can modify a reflex response via a neuron to the motor neuron of the reflex arc (e.g. to stop us dropping a hot object).	3. a) explain how the structure of a reflex arc, including the relay neuron, is related to its function b) describe practical investigations into reflex actions *PAG6*	
Research into the structure and function of the brain has huge potential impact for the ageing population. We know the brain is made of billions of neurons. We also know that different areas of the brain are important in different functions. **However, our ability to investigate and develop explanations about brain function remains limited. Most areas of the brain are concerned with many functions, but some functions can be mapped to particular areas using functional magnetic resonance imaging (fMRI), studies of patients with brain damage, and electrical stimulation (IaS3). There are ethical issues associated with studying brain damaged patients, including informed consent (IaS4).**	4. describe the structure and function of the brain and roles of the cerebral cortex (intelligence, memory, language and consciousness), cerebellum (conscious movement) and brain stem (regulation of heart and breathing rate) *(separate science only)* **5. explain some of the difficulties of investigating brain function** ***(separate science only)***	**Ideas about Science:** • **fMRI as a technological development that enabled observations that led to new scientific explanations (IaS3)** • **ethical issues around studying brain damaged patients (IaS4)**

B5.3 How do hormones control responses in the human body?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
The endocrine system of humans and other animals uses hormones, secreted by glands and transported by the blood, to enable the body to respond to external and internal stimuli. Hormones bind to receptors on effectors, stimulating a response. The endocrine system provides slower, longer-lasting responses than the nervous system. **The production of hormones is regulated by negative feedback.**	1. describe the principles of hormonal coordination and control by the human endocrine system **2. explain the roles of thyroxine and adrenaline in the body, including thyroxine as an example of a negative feedback system**	

B5.4 Why do we need to maintain a constant internal environment?	
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*
Cells, enzymes and life processes function only in certain conditions, and optimally when conditions are within a narrow range. The maintenance of a constant internal environment is homeostasis, and depends on receptors, nerves, hormones and (often antagonistic) effectors to counteract changes. The skin and muscles (including muscles in artery walls and hair follicles) work to control internal body temperature. **Maintenance of an ideal internal temperature depends on temperature receptors in the skin and hypothalamus, processing in the hypothalamus, responses such as sweating, hair erection, shivering, vasoconstriction and vasodilation, and negative feedback.** The kidneys filter water and urea from the blood into kidney tubules then reabsorb as much water as required, to maintain the water balance of the body. This helps to ensure the blood plasma is at the correct concentration to prevent the shrinking or bursting of cells due to osmosis. **Maintenance of an ideal water balance depends on receptors and processing in the hypothalamus, response by the pituitary gland (increased/decreased ADH production), and negative feedback.**	1. explain the importance of maintaining a constant internal environment in response to internal and external change 2. a) describe the function of the skin in the control of body temperature, including changes to sweating, hair erection and blood flow b) describe practical investigations into temperature control of the body *PAG6* *(separate science only)* **3. explain the response of the body to different temperature challenges, including receptors, processing, responses and negative feedback *(separate science only)*** 4. explain the effect on cells of osmotic changes in body fluids ⓘ *Learners are not expected to discuss water potential* *(separate science only)* 5. describe the function of the kidneys in maintaining the water balance of the body, including filtering water and urea from the blood into kidney tubules then reabsorbing as much water as required *(separate science only)* **6. describe the effect of ADH on the permeability of the kidney tubules *(separate science only)*** **7. explain the response of the body to different osmotic challenges, including receptors, processing, response, and negative feedback *(separate science only)*** 8. in the context of maintaining a constant internal environment: a) extract and interpret data from graphs, charts and tables M2c b) translate information between numerical and graphical forms M4a

Linked learning opportunities

Specification links:
- the effects of temperature on enzyme activity (B3.1)

Practical work:
- compare skin temperature and core body temperature under different conditions
- model the control of temperature by trying to keep a beaker of water at 40°C using just a Bunsen burner (single effector) compared to a Bunsen burner and ice (antagonistic effectors)

B5.5 What role do hormones play in human reproduction?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	***Linked learning opportunities***
Hormones play a vital role in enabling sexual reproduction in humans: they regulate the menstrual cycle, including ovulation, in adult females. Without this process, sexual reproduction would not be possible.	1. describe the role of hormones in human reproduction, including the control of the menstrual cycle	
A number of hormones interact to control the menstrual cycle: • **FSH causes the ovaries to develop a follicle containing an egg, and produce oestrogen** • **oestrogen causes the uterus wall to thicken** • **LH causes the follicle to release the egg (ovulation)** • **the remains of the follicle secretes progesterone** • **progesterone prepares the lining of the uterus for implantation of a fertilised egg** • **oestrogen and progesterone stop the production of LH and FSH** • **as progesterone levels fall, the thickened uterus wall breaks down and is discharged (menstruation).**	**2. explain the interactions of FSH, LH, oestrogen and progesterone in the control of the menstrual cycle**	
The menstrual cycle can be controlled artificially by the administration of hormones, often as an oral pill. The hormones prevent ovulation, so can be used as a contraceptive, but they do not decrease the risk of sexual transmission of communicable diseases (IaS4).	3. explain the use of hormones in contraception and evaluate hormonal and non-hormonal methods of contraception	**Specification links:** • sexually transmitted disease (B2.1) **Ideas about Science:** • risk in the context of sex and contraception (IaS4)
Hormones can also be used to artificially manipulate the menstrual cycle as a treatment in certain cases of female infertility in which follicle development and ovulation do not occur successfully. The use of hormones to treat infertility is an example of an application of science that has made a significant positive difference to people's lives (IaS4).	**4. explain the use of hormones in modern reproductive technologies to treat infertility**	**Ideas about Science:** • **infertility treatment as an application of science that makes a positive difference to lives (IaS4)**

B5.6 What can happen when organs and control systems stop working?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
Blood sugar level is controlled by insulin **and glucagon acting antagonistically**. Type 1 diabetes arises when the pancreas stops making insulin; blood sugar can be regulated using insulin injections. Type 2 diabetes develops when the body no longer responds to its own insulin or does not make enough insulin; blood sugar can be regulated using diet (high in complex carbohydrates), exercise and insulin injections.	1. explain how insulin controls the blood sugar level in the body **2. explain how glucagon and insulin work together to control the blood sugar level in the body** 3. compare type 1 and type 2 diabetes and explain how they can be treated	
The human eye is an important sense organ. Tissues in the eye are adapted to enable it to function. Damage or degradation of tissues and cells in the eye can impair sight. Ray diagrams can be used to model some of these problems and how they can be overcome (IaS3).	4. a) explain how the main structures of the eye are related to their functions, including the cornea, iris, lens, ciliary muscle and retina and to include the use of ray diagrams b) describe practical investigations into the response of the pupil in different light conditions *PAG6* *(separate science only)* 5. describe common defects of the eye, including short-sightedness, long-sightedness and cataracts, and explain how these problems may be overcome, including using ray diagrams to illustrate the effect of lenses *(separate science only)*	**Specification links**: • lenses and ray diagrams (P1.4) **Practical work**: • investigate the diameter of the pupil in different light conditions • investigate the focusing of light using lenses
Damage to neurons can lead to debilitating illness. Neurons, once differentiated, do not undergo mitosis, so cannot divide to replace lost neurons; this means damage to the nervous system can be difficult or impossible to treat. **Research into the use of stem cells to replace damaged cells of the nervous system offers the potential to improve the lives of people with nervous system injury and disease, but the ethics of stem cells use must also be considered (IaS4).**	**6. explain some of the limitations in treating damage and disease in the brain and other parts of the nervous system *(separate science only)***	**Specification links:** • Is it right to use stem cells in medicine? (B4.5) **Ideas about Science:** • stem cell therapy as an application of science that could change lives (IaS4)